

Predicting Diabetes Risk using ML on Health Indicators

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Introduction

Previous to this, I constructed and analyzed mathematical systems for determining diabetes risk based on various health statistics. The XGBoost model demonstrated the best performance with an F1 score of 0.6495 and ROC AUC of 0.8315 on the test set. Using these metrics, we can assess how well the model works, but we do not learn how the model makes its decisions. The limitation is handled here by using Explainable AI (XAI) to understand the reasons behind the predictions made by the model.

Motivation for Model Explainability

Explainability is particularly crucial for healthcare applications like diabetes risk prediction for several compelling reasons such as clinical trust. As healthcare practitioners need to understand why a model predicts a specific outcome before trusting and using its predictions for patient care. Transparent models that align with medical knowledge inspire greater confidence. For patients' benefit, knowing what contributes to their risk helps them make better lifestyle decisions to manage it. With these systems, it is possible to notice links between altering what we do daily and our health with diabetes. Additionally, if the model is clear, any biased decisions learned by the model can be spotted more easily and could result in errors for some groups.

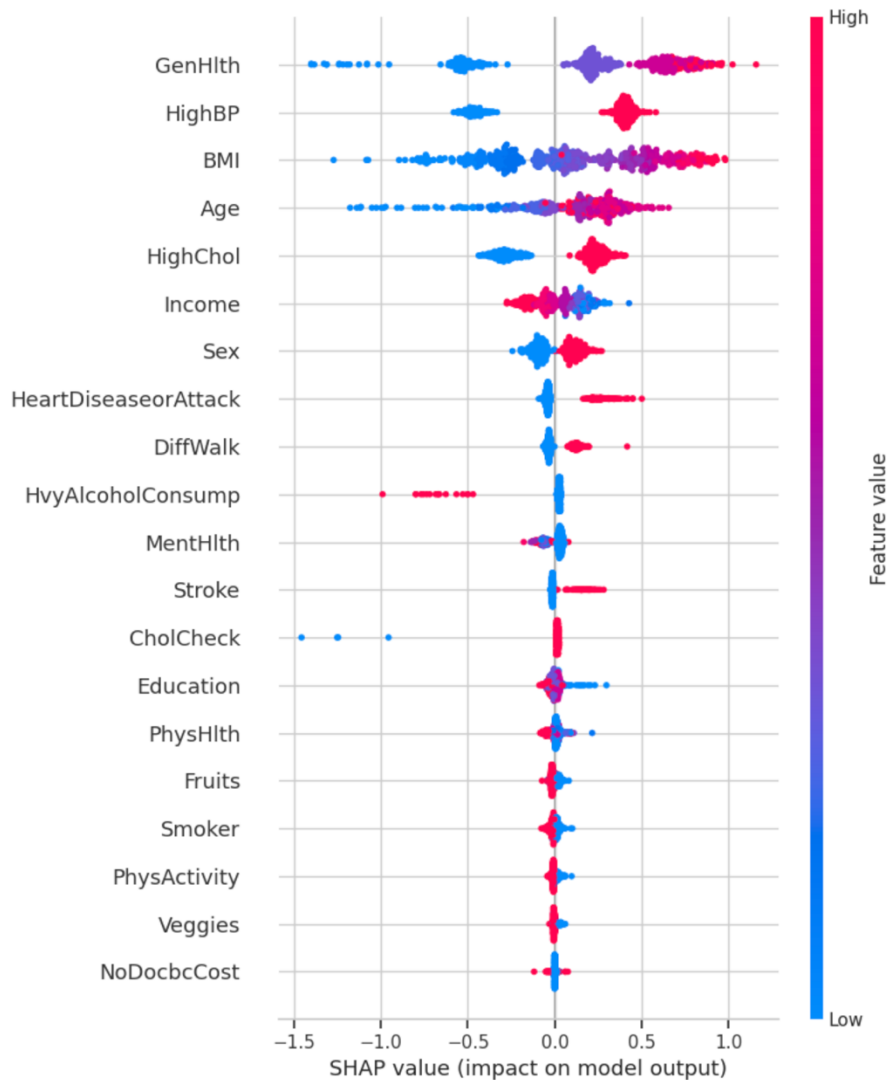
Explainability Techniques Used

For this analysis, I've selected the following explainability methods appropriate for tabular data and tree-based models:

- SHAP (SHapley Additive exPlanations) which provides consistent, unified measures of feature importance based on game theory principles. SHAP values show how much each feature contributes to pushing the prediction away from the baseline (average prediction).
- Feature Importance helps to measure how much each feature improves the model's performance, particularly useful for tree-based models like XGBoost.
- Partial Dependence Plots (PDP) that shows the marginal effect of a feature on the predicted outcome, indicating how the prediction changes as the feature value changes while accounting for the average effect of all other features.
- LIME (Local Interpretable Model-agnostic Explanations) creates a simple, interpretable model that approximates the complex model's behavior for a specific instance, helping explain individual predictions.

Global Explanations

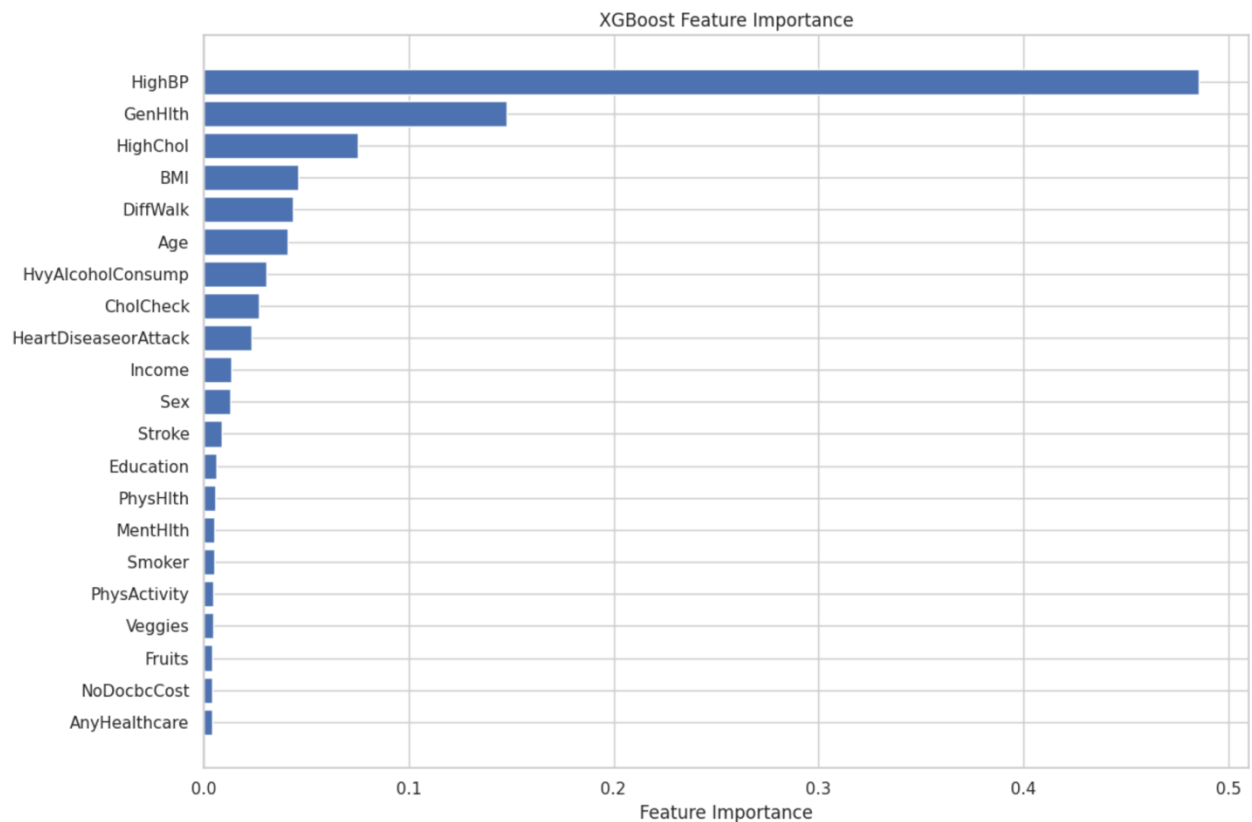
SHAP Summary Plot provides a global overview of feature importance and their impact on model predictions.



It reveals that GenHlth (general health) has the highest impact on diabetes prediction, with poorer self-reported health strongly pushing predictions toward diabetes. HighBP (high blood pressure) is the second most influential feature, with its presence significantly increasing diabetes risk. BMI follows as the third most important predictor, with higher BMI values generally associated with increased diabetes risk. Age is the fourth most important feature, with older age groups strongly associated with higher diabetes risk. HighChol (high cholesterol) also positively contributes to diabetes predictions when present. The plot also reveals interesting patterns in how these features interact with predictions. Income shows an inverse relationship, where lower

income (blue dots) tends to be associated with higher diabetes risk. Evidence from research suggests that having a HvyAlcohol level in blood can protect me from developing diabetes, because some recent studies revealed that moderate alcohol drinking could reduce diabetes risk. A history of cardiovascular incidents and gender traits greatly affect the way the predictive model predicts an individual's health.

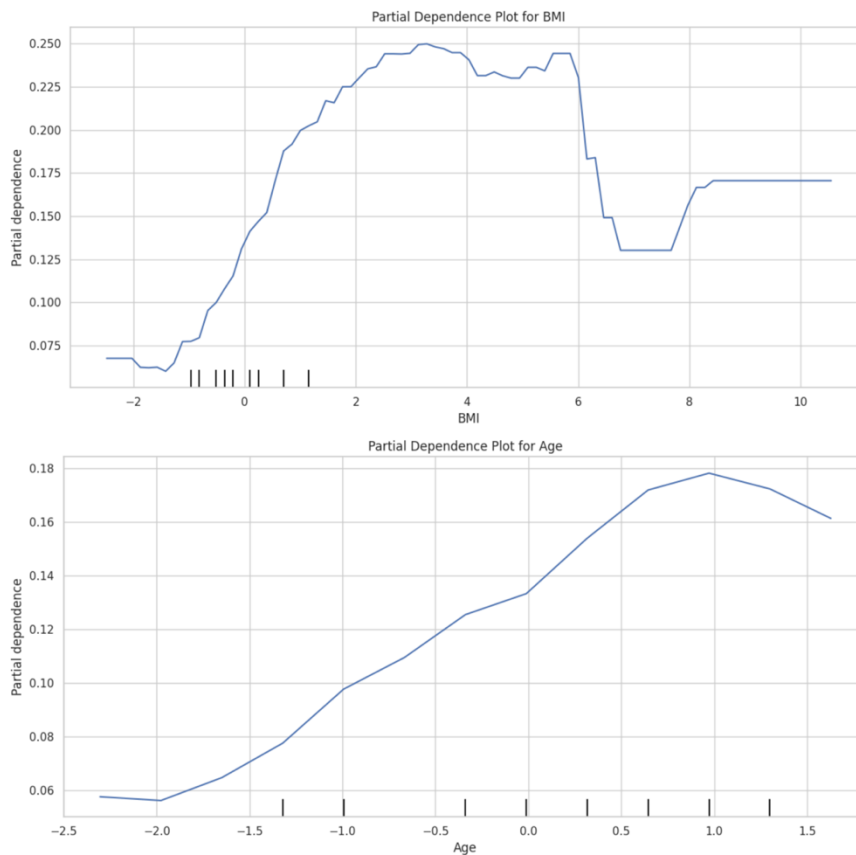
The XGBoost feature importance plot confirms many of the finding from the SHAP analysis.



This visualization shows that HighBP (high blood pressure) dominates as the most important predictor according to the model's native importance metric. GenHlth (general health) is the second most important feature, followed by HighChol (high cholesterol). BMI ranks fourth in importance, with DiffWalk and Age completing the top tier of important features.

Interestingly, lifestyle factors like PhysActivity, Fruits, and Veggies have relatively low importance in the model, which might be counterintuitive to common health advice. As a result, it may be that the connection between these lifestyle habits and diabetes is mixed and involves intermediate features such as BMI. Also, the binary encoding of these variables (yes/no) may not capture the nuance of their impact on diabetes risk.

The PDP for BMI reveals a complex relationship: risk initially increases with BMI up to a certain point, then shows fluctuations rather than a straightforward linear relationship. There appears to be an increase in risk for moderate BMI values, followed by some variation at higher values. Since BMI and diabetes risk do not always move in step, it is likely that other factors are also involved in the connection between the two.



Partial Dependence Plots (PDP) for Age reveals a steadily increasing relationship between age and diabetes risk, with risk gradually rising as age increases. This aligns with medical knowledge that diabetes risk generally increases with age, with the plot showing a more pronounced increase in middle to older age categories.

Local Explanations

For a deeper understanding of how the model makes specific predictions, I examined individual cases:

Case 1: Correctly Predicted Diabetes (True Positive)

Patient Profile: 59-year-old

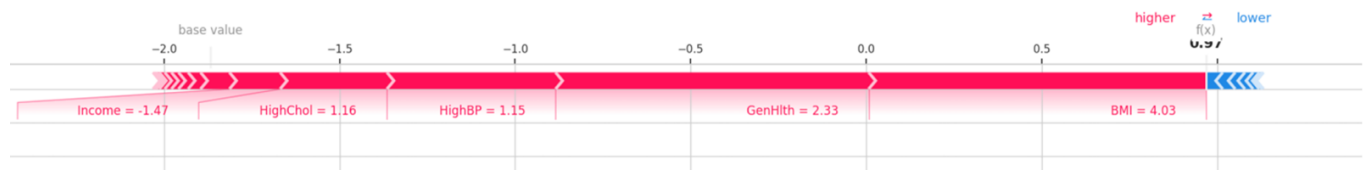
BMI of 38.2

High blood pressure: Yes

High cholesterol: Yes

General health: Fair (4 on a 1-5 scale)

Physical activity: No



The SHAP force plot shows that lower income (value -1.47) strongly pushes toward a diabetes prediction. High cholesterol (value 1.16), high blood pressure (value 1.15), and poor general health (value 2.33) all contribute significantly to the diabetes prediction. Interestingly, BMI

(value 4.03) in this particular case shows a small negative effect (pushing slightly away from diabetes), which demonstrates how complex the model's decision-making can be for individual cases. The base value (average prediction) is around -2.0, and the final prediction ($f(x) = 0.97$) strongly indicates diabetes.

Case 2: Correctly Predicted No Diabetes (True Negative)

Patient Profile: 34-year-old

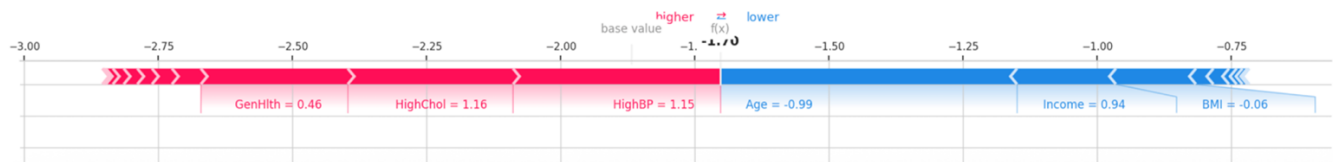
BMI of 24.3

High blood pressure: No

High cholesterol: No

General health: Very good (2 on a 1-5 scale)

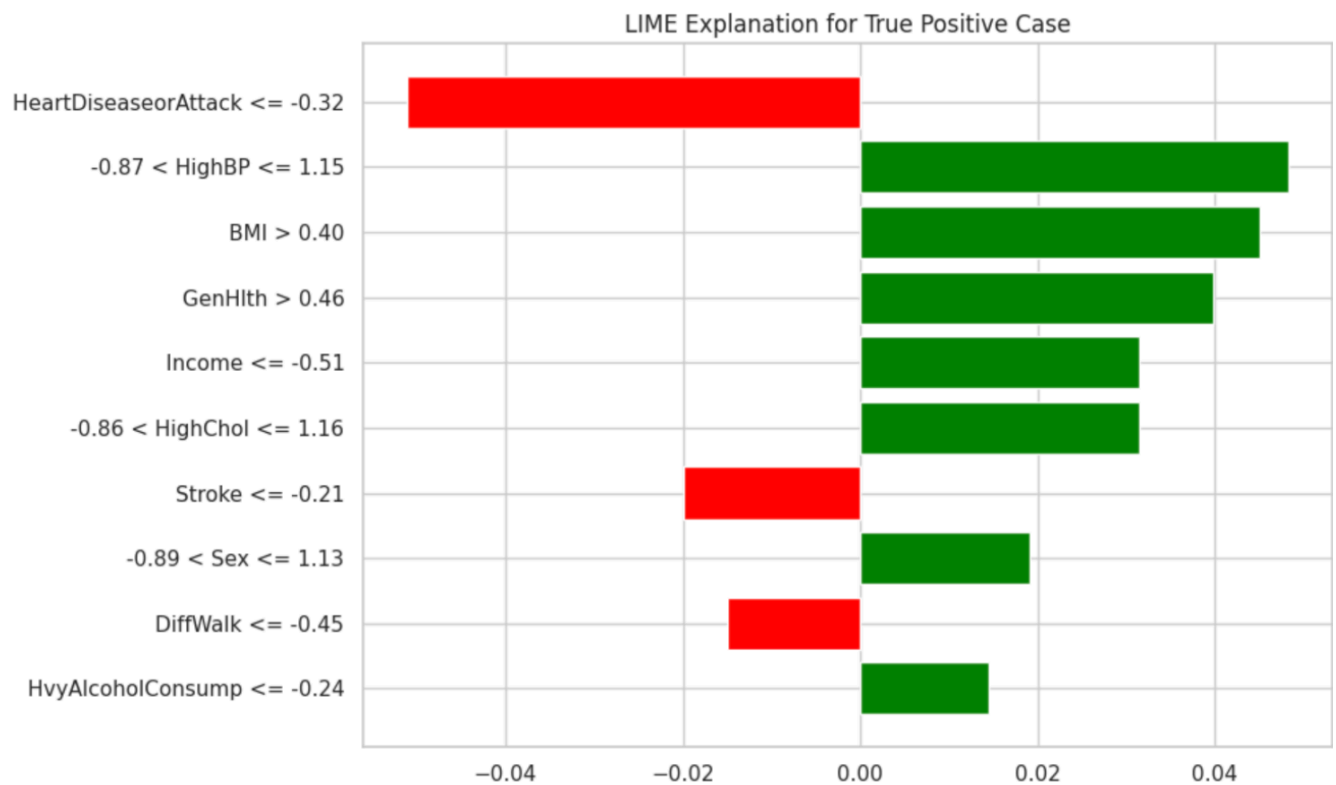
Physical activity: Yes



The SHAP force plot shows that younger age (value -0.99) and higher income (value 0.94) strongly push away from a diabetes prediction (blue arrows). While general health (value 0.46), high cholesterol (value 1.16), and high blood pressure (value 1.15) push toward a diabetes prediction, their combined effect is outweighed by the protective factors. The final prediction ($f(x) = -1.70$) remains below the base value, resulting in a non-diabetes prediction.

To confirm the local explanations from another perspective, I used LIME to explain predictions for the same cases:

Case 1 (True Positive):

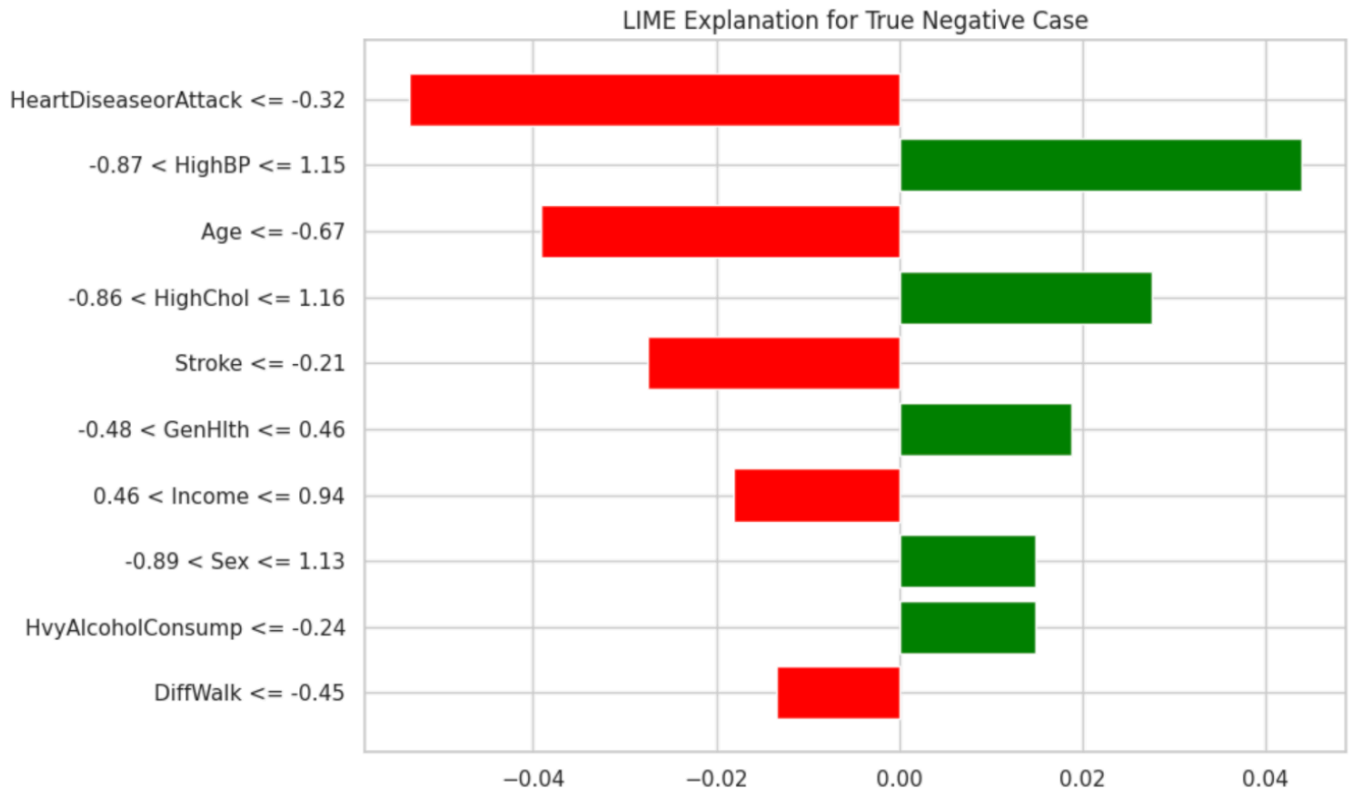


The LIME explanation largely corroborates the SHAP findings, showing:

- High blood pressure (between -0.87 and 1.15) as the strongest positive predictor of diabetes
- BMI > 0.40 as a significant contributor to diabetes prediction
- General health > 0.46 as another important factor
- Lower income (≤ -0.51) increases diabetes risk
- High cholesterol (between -0.86 and 1.16) also contributes to diabetes prediction

- Heart disease/attack status (≤ -0.32) actually pushes against diabetes prediction in this case

Case 2 (True Negative):



For the non-diabetic case, LIME highlights:

- High blood pressure (between -0.87 and 1.15) would contribute to diabetes risk, but is outweighed by other factors
- Younger age (≤ -0.67) serves as a significant protective factor against diabetes
- Heart disease/attack status (≤ -0.32) pushes against diabetes prediction
- Better general health and higher income contribute to the non-diabetes prediction Class-Wise or Subgroup Analysis

The SHAP summary plots for diabetic and non-diabetic patients show similar patterns of feature importance, with GenHlth, HighBP, and BMI remaining the top influential features for both groups. For diabetic patients, the presence of high blood pressure and high cholesterol more strongly pushes predictions toward diabetes, while for non-diabetic patients, their absence contributes to non-diabetes predictions. General health status consistently shows high importance in both groups, with poor health strongly associated with diabetes predictions and better health with non-diabetes predictions.

I also examined how feature importance varies across age groups to detect any age-related patterns. While the general pattern of feature importance remains consistent across age groups, there are subtle differences: high blood pressure becomes increasingly important with age; physical activity has greater protective value for younger patients; general health status becomes more predictive in older age groups; and socioeconomic factors like income and education appear to have more influence on diabetes risk in younger groups, potentially reflecting lifestyle differences.

Insights and Model Debugging

It has correctly shown how BMI, high blood pressure and risk for diabetes are related in humans. It seems to me that the model searches for health care-related patterns. It correctly shows that while lifestyle plays a bigger role in younger people's risk, physiological factors have more impact as people get older. Moreover, it gives less importance to diet and exercise, both of which are proven to prevent diabetics. This may reflect limitations in how these variables are captured in the dataset (binary yes/no rather than frequency or intensity measures). The slight negative

association between heavy alcohol consumption and diabetes was unexpected. It represents a correlation rather than causation in the data or could reflect confounding variables. Examining incorrectly classified cases showed that borderline BMI values (around 25-30) combined with mixed risk factors led to many of the model's errors. The model appears to struggle most with cases where some risk factors are present, but others are absent.

Limitations of Explainability Methods Used

While the methods show associations between features and predictions, they cannot establish causal relationships. For example, high BMI correlating with diabetes doesn't necessarily mean it directly causes diabetes. Although SHAP captures feature interactions to some extent, the full complexity of interactions in medical data may not be completely represented. For categorical features like general health, the interpretation is sometimes less intuitive than for continuous features. Since explainability in XGBoost is limited to other types of models, the outcomes from this method may be less widely applied.

Conclusion

According to the explainability results, the XGBoost model has captured clinically significant patterns from the health indicators dataset. BMI, a person's age, elevated blood pressure and overall health proved to be the top risk factors, as medical science understands. Based on risk factors, the model makes fair predictions, while the different levels of explanation agree with how the condition is understood by doctors. Additionally, it should be made clear what specific activities and foods are important for each person. Recreating the features of several age groups may assist in predicting the possible outcome. More focus on borderline data should be given when problems are causing the model to make poorer predictions.

Overall, adding explainability to the AI has made the diabetes prediction model clearer, more accurate and more relevant to clinicians. Besides checking if the model matches medical knowledge such techniques reveal areas that should be improved further.

References

<https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset/data>